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An Introduction to Chemical Kinetics

Claire Vallance

Chapter 7

Chain reactions and explosions

7.1 Chain reactions

Chain reactions are complex reactions that involve *chain carriers*, the name given to reactive intermediates that react to produce further reactive intermediates. Some examples of chain reactions are:

- combustion of a fuel gas;
- development of rancidity in fats;
- the polymerase chain reaction (PCR) used to amplify DNA samples for analysis;
- many polymerisation reactions, e.g. polymerisation of ethene to polyethene;
- nuclear fission caused by neutron bombardment.

The elementary steps in a chain reaction may be classified into *initiation*, *propagation*, *inhibition*, and *termination* steps. As an example, consider the infamous reaction by which chlorofluorocarbons (CFCs) destroy ozone:

$C_nF_mCl + h\nu \longrightarrow C_nF_m + Cl$	Initiation
$Cl + O_3 \longrightarrow ClO + O_2$	Propagation
$ClO + O \longrightarrow Cl + O_2$	Propagation
$Cl + CH_4 \longrightarrow CH_3 + HCl$	Termination

The various steps in the mechanism are classified as follows:

1. *Initiation step*
   The reaction is initiated either thermally or photochemically. The first reactive intermediates/chain carriers (in this case a Cl radical) are produced.

2. *Propagation step*
   The reaction of a radical leads to the formation of another radical, i.e. another reactive intermediate. In the first propagation step above, Cl reacts to form ClO; in the second step ClO reacts to form Cl.

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3. *Termination step*
   Chain carriers are deactivated. Often this occurs through radical–radical recombination, reactions with walls, or reactions with another molecule to create an inactive product. Note that termination products may go on to be involved in other reactions, but are not involved further in the chain reaction of interest.

Some chain reactions also involve inhibition steps, in which product molecules are destroyed. Inhibition steps are sometimes also referred to as *retardation* or *de-propagation* steps.

The *chain length* in a chain reaction is defined as the number of propagation steps per initiation step, or alternatively as the rate of propagation divided by the rate of initiation. Chain lengths can be very long: in the above example a single Cl radical can destroy around 10^6 molecules of ozone

$$\text{chain length}, n = \frac{\text{rate of propagation}}{\text{rate of initiation}} = \frac{\text{rate of propagation}}{\text{rate of termination}}. \tag{7.1}$$

The reaction of Cl atoms with ozone is an example of a *cyclic chain reaction*. Atomic chlorine acts as a catalyst and is continuously regenerated until it is removed by a termination step. It is also possible to have *non-cyclic chain reactions*, involving many reactive species and elementary steps. Non-cyclic chain reactions can have extremely complicated kinetic mechanisms.

Chain reactions in which each propagation step produces only one reactive intermediate are called *linear chain reactions*. *Branched chain reactions* are also possible, in which a chain carrier reacts to form more than one chain carrier in a single elementary step. We will look at some examples of both linear and branched chain reactions in the following.

7.2 Linear chain reactions

The reaction between H_2 and Br_2 has become the 'benchmark' system for illustrating the kinetics of linear chain reactions, and we will use this reaction as our main example. While the reaction is not the most up-to-date of examples, having first been studied around 100 years ago, it provides a very clear illustration of all of the key features of chain reactions. Once we have studied the hydrogen-bromine reaction, we can also investigate the reactions of H_2 with Cl_2 and I_2. While we might expect the three reactions to have identical mechanisms, we in fact find them to be markedly different, providing an opportunity to investigate a variety of factors that determine the most favoured (i.e. fastest) mechanism.

7.2.1 The hydrogen-bromine reaction

As noted above, the kinetics of the reaction between H_2 and Br_2 were determined experimentally by Bodenstein and Lind around 100 years ago[1]. The overall reaction equation is

$$H_2 + Br_2 \longrightarrow 2HBr \tag{7.2}$$

and Bodenstein initially determined the rate law to be

$$\frac{d[HBr]}{dt} = k[H_2][Br_2]^{1/2}. \tag{7.3}$$

The measured order of 1/2 with respect to Br_2 indicated that the reaction proceeds via a complex reaction mechanism, rather than a simple bimolecular collision. Further investigation showed that this rate law in fact only holds for the early stages of the reaction, and that the true rate law takes the form

$$\frac{d[HBr]}{dt} = \frac{k[H_2][Br_2]^{1/2}}{1 + k'[HBr]/[Br_2]}. \tag{7.4}$$

Any proposed mechanism for the reaction must agree with both of these observations.

The first step in any chain reaction is the initiation step. The reaction between H_2 and Br_2 can be initiated by either thermally-induced or photon-induced dissociation of Br_2

$$Br_2 + M \longrightarrow Br + Br + M \tag{7.5}$$

or

$$Br_2 + h\nu \longrightarrow Br + Br. \tag{7.6}$$

We will concentrate on the thermal mechanism for the purposes of deriving a rate law for the overall reaction, but the steps following the initiation step are the same for both cases. The currently accepted mechanism is:

$Br_2 + M \xrightarrow{k_1} Br + Br + M$	Initiation
$Br + H_2 \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} H + HBr$	Propagation/inhibition
$H + Br_2 \xrightarrow{k_3} Br + HBr$	Propagation
$Br + Br + M \xrightarrow{k_4} Br_2 + M$	Termination

In the second step, because the H–H bond is stronger than the H–Br bond, the reverse (inhibition) reaction becomes possible once an appreciable amount of HBr has built up. The reaction chain contains two radical chain carriers, H and Br. In order to arrive at an overall rate law for the reaction, we apply the steady-state approximation to the two chain carriers.

[1] M Bodenstein and C Lind 1906 *Z. Phys. Chem.*, **57** 168.

$$\begin{aligned}\frac{d[\mathrm{H}]}{dt} &= 0 = k_2[\mathrm{Br_2}][\mathrm{H_2}] - k_{-2}[\mathrm{H}][\mathrm{Br}] - k_3[\mathrm{H}][\mathrm{Br_2}] \\ \frac{d[\mathrm{Br}]}{dt} &= 0 = 2k_1[\mathrm{Br_2}][\mathrm{M}] - k_2[\mathrm{Br}][\mathrm{H_2}] + k_{-2}[\mathrm{H}][\mathrm{HBr}] + k_3[\mathrm{H}][\mathrm{Br_2}] \\ &\quad - 2k_4[\mathrm{Br}]^2[\mathrm{M}].\end{aligned} \tag{7.7}$$

We can solve these two equations to obtain expressions for the concentrations of H and Br in terms of the reactant and product concentrations and the various rate constants. This is a good example of a case in which we can simplify the required algebra considerably through careful inspection. The two equations each depend on both chain carrier concentrations, and also share terms. We can simplify the process of solving the equations by adding them together to give:

$$0 = 2k_1[\mathrm{Br_2}][\mathrm{M}] - 2k_4[\mathrm{Br}]^2[\mathrm{M}]. \tag{7.8}$$

This expression can be rearranged to give the steady-state concentration of atomic bromine.

$$[\mathrm{Br}] = \left(\frac{k_1[\mathrm{Br_2}]}{k_4}\right)^{1/2}. \tag{7.9}$$

Note that equation (7.8) implies that the rate of initiation is the same as the rate of termination, as expected under steady-state conditions. This provides a good check that we have made no mistakes up to this point. Equation (7.8) also leads to a considerable simplification of the steady-state equation describing the rate of change of [Br], which now becomes

$$0 = -k_2[\mathrm{Br}][\mathrm{H_2}] + k_{-2}[\mathrm{H}][\mathrm{HBr}] + k_3[\mathrm{H}][\mathrm{Br_2}] \tag{7.10}$$

and may be rearranged to give an expression for the steady-state concentration of H atoms.

$$\begin{aligned}[\mathrm{H}] &= \frac{k_2[\mathrm{Br}][\mathrm{H_2}]}{k_2[\mathrm{HBr}] + k_3[\mathrm{Br_2}]} \\ &= \frac{k_2[\mathrm{H_2}]}{k_2[\mathrm{HBr}] + k_3[\mathrm{Br_2}]}\left(\frac{k_1[\mathrm{Br_2}]}{k_4}\right)^{1/2}\end{aligned} \tag{7.11}$$

where in the second step above we have substituted in our previously determined expression (equation (7.9)) for the concentration of Br. Now that we have determined the concentrations of the reactive intermediates, we are ready to determine the overall reaction rate. This is found from the rate of change of the concentration of the HBr product. From the reaction mechanism, we have

$$\frac{d[\mathrm{HBr}]}{dt} = k_2[\mathrm{Br}][\mathrm{H_2}] - k_{-2}[\mathrm{H}][\mathrm{HBr}] + k_3[\mathrm{H}][\mathrm{Br_2}]. \tag{7.12}$$

Substituting in our expressions for [H] and [Br] (equations (7.11) and (7.9)) gives

$$\frac{d[\mathrm{HBr}]}{dt} = \frac{2k_2(k_1/k_4)^{1/2}[\mathrm{Br_2}]^{1/2}[\mathrm{H_2}]}{1 + (k_{-2}/k_3)[\mathrm{HBr}]/[\mathrm{Br_2}]}. \tag{7.13}$$

We see that this agrees with the measured rate law, equation (7.4). In the early stages of the reaction, the concentration of the HBr product is much lower than that of the reactant Br_2, and the second term in the denominator becomes negligible. The rate law then reduces to

$$\frac{d[\mathrm{HBr}]}{dt} = 2k_2(k_1/k_4)^{1/2}[\mathrm{Br_2}]^{1/2}[\mathrm{H_2}] \tag{7.14}$$

again reproducing the experimental observations. The proposed mechanism therefore fits well with the experimental measurements.

7.2.2 The hydrogen–chlorine reaction

The mechanism for the hydrogen–chlorine reaction is similar to that for the hydrogen–bromine reaction

$Cl_2 + M \xrightarrow{k_1} Cl + Cl + M$	Initiation
$Cl + H_2 \xrightarrow{k_2} H + HCl$	Propagation
$H + Cl_2 \xrightarrow{k_3} Cl + HCl$	Propagation
$Cl + Cl + M \xrightarrow{k_4} Cl_2 + M$	Termination

As in the case of the $H_2 + Br_2$ reaction, the reaction may also be initiated photochemically. However, unlike the $H_2 + Br_2$ reaction, both propagation steps are very efficient, and the inhibition step is so slow that we have omitted it from the mechanism. In combination, these factors result in the rate for the $H_2 + Cl_2$ reaction being much higher than that for $H_2 + Br_2$. Chain lengths up to 10^6 are possible, and, coupled with the exothermicity of the reaction, these can lead to a thermal explosion (for more on explosions, see section 7.4).

The $H_2 + Cl_2$ reaction provides a good example of a case for which the steady-state approximation breaks down. Since both propagation steps are very efficient, when the radical concentrations [H] and [Cl] are much lower than the reactant concentrations $[H_2]$ and $[Cl_2]$, reactive collisions between a Cl atom and an H_2 molecule (propagation) are much more likely than terminating collisions of Cl with another Cl atom. This means that the reaction may be very advanced before the steady-state condition—that the rates of initiation and termination are equal—is reached. In practice, the situation is usually simplified somewhat, due to the extreme sensitivity of the reaction to inhibition by contaminants, such as O_2. Molecular oxygen reacts rapidly with H and Cl radicals to form inert radicals, providing a number of alternative termination pathways and increasing the overall rate of termination. If O_2 is present at a concentration of around 1% or greater, the following termination steps dominate to the point where the steady-state approximation becomes valid

$$\begin{aligned} H + O_2 + M &\xrightarrow{k_5} HO_2 + M \\ Cl + O_2 + M &\xrightarrow{k_6} ClO_2 + M. \end{aligned} \tag{7.15}$$

If we replace the termination step (step 4) of the original reaction mechanism with the termination steps above, we can apply the steady-state approximation to obtain expressions for the steady state H and Cl concentrations and therefore (after some algebra, left as an exercise for the keen reader) for the overall rate

$$\frac{1}{2}\frac{d[HCl]}{dt} = \frac{2k_a[H_2][Cl_2]^{1/2}}{[O_2]([H_2] + k_b[Cl_2])} \tag{7.16}$$

with

$$k_a = \frac{k_1 k_2}{k_5} \tag{7.17}$$

and

$$k_b = \frac{k_3 k_6}{k_2 k_5}. \tag{7.18}$$

7.2.3 The hydrogen–iodine reaction

We might expect the $H_2 + I_2$ reaction to have a similar mechanism to the Br_2 and Cl_2 analogues. However, the second step in the mechanism, $I + H_2 \longrightarrow H + HI$, occurs much too slowly at normal temperatures for this mechanism to be viable. Various kinetic mechanisms operate at different temperatures; for example,

$$\begin{aligned} I_2 + M &\overset{K}{\rightleftharpoons} I + I + M && \text{Pre-equilibrium} \\ I + I + H_2 &\xrightarrow{k_2} 2HI && \text{Termolecular reaction} \end{aligned}$$

Unlike the complicated rate laws followed by the chlorine and bromine reactions, the mechanism above gives rise to a simple second order rate law (this can be proved as an exercise using the pre-equilibrium treated in section 5.2 as an example)

$$\frac{d[HI]}{dt} = 2k_2K[I_2][H_2]. \tag{7.19}$$

7.3 Comparison between the hydrogen–halogen reactions

The key difference between the reactions of Cl_2, Br_2 and I_2 with molecular hydrogen lies in the relative endothermicity of the reactions between the halogen atoms and H_2 (step 2 in the chain reaction mechanisms)

$$\begin{aligned} Cl + H_2 &\longrightarrow HCl + H && \Delta H = 4.4 \text{ kJ mol}^{-1} \\ Br + H_2 &\longrightarrow HBr + H && \Delta H = 69.6 \text{ kJ mol}^{-1} \\ I + H_2 &\longrightarrow HI + H && \Delta H = 137.7 \text{ kJ mol}^{-1} \end{aligned}$$

The iodine reaction is highly endothermic, and consequently so slow that it does not occur at accessible temperatures, ruling out the chain mechanism observed for the other two halogens. The chlorine reaction is almost thermoneutral and, therefore, fast, and the bromine reaction is relatively endothermic and slow. While the reverse (inhibition) step $H + HX \longrightarrow X + H_2$ is thermodynamically favourable for both Cl and Br, it is unimportant for Cl, due to the very low steady-state concentration of H atoms. In contrast, the inhibition step has a considerable effect on the overall kinetics of the bromine reaction.

7.4 Explosions and branched chain reactions

Explosions without doubt represent one of the most exciting phenomena in chemistry, and often play a significant role in encouraging an early interest in chemistry amongst aspiring scientists. However, for the most part, professional chemists and chemical engineers are rather keen not to experience explosions as they go about their daily business, and it is therefore very important to understand the conditions under which a reaction mixture can become explosive, in order that such conditions can be avoided if at all possible.

An explosion occurs when a chemical reaction accelerates out of control. As the reaction speeds up, gaseous products are formed in larger and larger amounts, and more and more heat is generated. The rapid liberation of heat causes the gases to expand, generating extremely high pressures, and it is this sudden formation of a huge volume of expanded gas that constitutes the explosion. The resulting pressure wave travels at very high speeds, often much faster than the speed of sound, generating a supersonic shock wave or 'bang'.

There are two different mechanisms that may lead to an explosion. These are related to the fact that the overall reaction rate depends on both the magnitude of the rate constant and the amounts of reactants present in the reaction mixture.

If the heat generated in a reaction due to the reaction exothermicity cannot be dissipated sufficiently rapidly, the temperature of the reaction mixture increases. This increases the rate constant and, therefore, the reaction rate, producing more heat and accelerating the reaction rate still further, and so the cycle continues until an explosion results. Such explosions are known as *thermal explosions*, and in principle may occur whenever the rate of heat production by a reaction mixture exceeds the rate of heat loss to the surroundings—often the walls of the reaction vessel. This type of explosion is unfortunately relatively common when a chemist 'scales up' a reaction that has previously been carried out in small quantities, without properly taking into account the laws of thermodynamics.

The second category of explosion arises from chain branching within a chain reaction. Explosions in this category are known as *chain branching explosions*, or sometimes, somewhat misleadingly, *isothermal explosions*. In this case, one or more steps in the reaction mechanism produce two or more chain carriers from a single chain carrier. As the reaction proceeds, the number of chain carriers increases exponentially, leading to a rapid acceleration in the overall reaction rate.

In practice, both mechanisms often occur simultaneously, since any acceleration in the rate of an exothermic reaction due to chain branching will eventually lead to

an increase in temperature. It is very important to note, however, that chain branching is not a requirement for an explosion. As an example, the detonation of TNT (2,4,6-trinitrotoluene) is simply the result of an extremely fast chemical decomposition that generates huge quantities of gas.

The reaction $2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(g)}$ provides an example of a reaction in which both mechanisms are important, and we will consider its mechanism in some detail.

7.4.1 The reaction $2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(g)}$

Though the reaction between molecular oxygen and molecular hydrogen may appear to be very simple, the mechanism is in fact extremely complex, and is still not fully understood. The process is known to be a branched chain reaction, involving carriers such as H, O, and OH. A simplified version of the thermal mechanism is given below

$H_2 + \text{wall} \xrightarrow{k_1} H + H + \text{wall}$	Initiation
$H + O_2 \xrightarrow{k_2} OH + O$	Branching
$O + H_2 \xrightarrow{k_3} OH + H$	Branching
$OH + H_2 \xrightarrow{k_4} H + H_2O$	Propagation
$H + O_2 + M \xrightarrow{k_5} HO_2 + M$	Termination
$H + \text{wall} \xrightarrow{k_6} \text{H-wall}$	Termination
$O + \text{wall} \xrightarrow{k_7} \text{O-wall}$	Termination
$OH + \text{wall} \xrightarrow{k_8} \text{OH-wall}$	Termination

Reaction steps 2 and 3 in the mechanism both produce two radical chain carriers for each chain carrier consumed. The two steps combine to give an overall branching coefficient of three, corresponding to the hypothetical reaction $H + O_2 + H_2 \longrightarrow H + OH + OH$. Even so, the mixture is not explosive under all conditions, mainly because steps 1, 2, and 3 are endothermic ($\Delta H_1 = 427$ kJ mol^{-1}, $\Delta H_2 = 71$ kJ mol^{-1}, $\Delta H_3 = 17$ kJ mol^{-1}) and are therefore slow at low temperatures. The efficiency of the branching steps increases with increasing temperature, and as a result, the reaction displays a complex dependence on temperature and pressure, as shown in figure 7.1. We will explore this dependence in the following.

Figure 7.1 shows the regions of temperature and pressure over which the reaction is explosive (shaded regions) or proceeds steadily without exploding (unshaded regions). Consider the vertical dotted line on the diagram. Travelling vertically from the bottom to the top of the line corresponds to gradually increasing the pressure at a fixed temperature of 800 K. We see that at low pressures the reaction proceeds steadily, then reaches a range of pressures over which it becomes explosive, before returning to steady reaction at higher pressures, and finally becoming explosive again at very high pressures. The regions are demarcated by a number of *explosion limits*, also marked on the diagram. It is relatively straightforward to explain the presence of the various explosion limits by reference to the reaction mechanism.

1. At very low pressures the mean free path in the gas is large, and chain carriers can reach the walls and combine. Collisions with walls are more

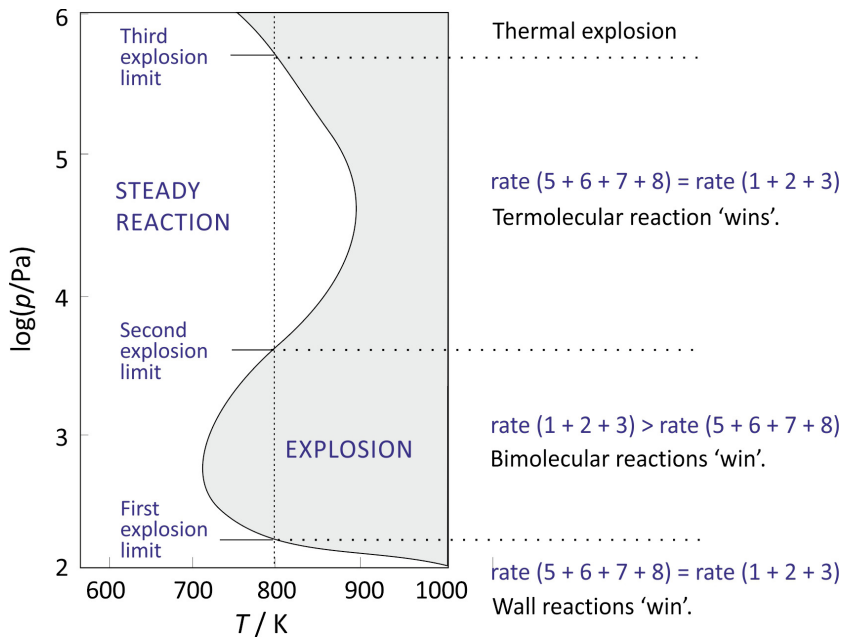

Figure 7.1. Explosion diagram for the reaction between H_2 and O_2.

likely than collisions with other gas-phase molecules, so that overall the rate of steps 5, 6, 7, and 8 balances that of steps 1, 2, and 3, i.e. termination balances initiation, and steady reaction occurs.

2. At higher pressures, the chain carriers react before reaching the walls, and the gas-phase branching ratios become too high for wall termination to control. This is primarily due to the fact that the rates of the bimolecular gas-phase reactions are proportional to $(\text{pressure})^2$, while the rate of wall reactions increases in direct proportion to the pressure. As the pressure increases, the rate of reactions 1, 2, and 3 exceeds that of reactions 5, 6, 7, and 8, and the mixture becomes explosive once the pressure reaches the first explosion limit.

3. At even higher pressures, three body collisions start to become important. Termination step 5, whose rate is proportional to $(\text{pressure})^3$, can now match steps 1 to 3 in efficiency. The collective rate of reactions 5, 6, 7, and 8 now outweighs the rate of steps 1, 2, and 3, leading to steady reaction at pressures above the second explosion limit.

4. If the pressure is increased still further, the reaction rate increases so much, and such a large amount of heat is generated, that a thermal explosion results above the third explosion limit.

We can also see from figure 7.1 that if we change the temperature, the explosion limits occur at different pressures. Other factors, such as the shape and size of the reaction vessel, and the presence of an inert gas, can also change the explosion limits.

These effects are fairly straightforward to understand if we consider the likely effect of each variable on the relative rates of initiation, propagation/branching, and termination steps.

1. *Temperature*
   Increasing the temperature increases the efficiency both of endothermic reaction steps and steps for which there is an activation barrier. This applies to steps 1–4 in the above mechanism. The termination steps are less sensitive to temperature, and may even be slowed down since they tend to be exothermic. As a result, the first explosion limit is lowered, as the rates of steps 2 and 3 outpace those of steps 6–8 more readily. The second explosion limit is increased because at higher temperatures a higher pressure is needed in order for the termolecular step 5 to become important. The third limit is decreased, since at higher temperatures more heat is produced, and also the heat that is produced is harder to lose from the system.

2. *Surface-to-volume ratio*
   The shape and size of the reaction vessel can have a considerable effect on the explosion limits. Increasing the surface-to-volume ratio favours processes that involve collisions with the vessel walls over gas phase processes, which in this case means the initiation step 1 and termination steps 6, 7, and 8. The high efficiency of the branching steps means that step 1 is unimportant in determining the overall rate, and therefore the increased efficiency of the termination steps increases the pressure at which the first explosion limit is reached. The second limit has no dependence on the vessel walls and is unchanged. The third limit increases because it becomes easier to lose heat from the system due to the greater number of collisions with the walls.

3. *Total pressure*
   Adding an inert gas to the mixture decreases the mean free path of the gas molecules, and disfavours collisions with the walls. This lowers the first explosion limit, since the termination steps 6, 7, and 8 become less efficient. The second limit is also decreased because the inert gas can act as the third body M, increasing the rate of step 5, a termination process. The third limit is lowered due to the reduced heat transfer from the gas to the vessel walls.

We have seen that a number of different variables come into play when determining whether or not a reaction mixture is likely to become explosive. These should all be considered carefully when working with any potentially explosive reactions in order to minimise any safety risks.

7.5 Concluding remarks

Over the course of this short introduction to chemical kinetics, we have explored in some detail the concepts of elementary and complex reactions, defined the rate of reaction, and considered the various factors on which the rate depends. We have shown that the reaction rate may be related to reactant concentrations through the relevant

rate law, and to temperature through the Arrhenius equation. We have looked at a number of experimental techniques for measuring reaction rates, and a variety of analysis methods that allow us to determine the rate law from experimental data. Finally, we have explored the link between the rate law and the reaction mechanism, and have demonstrated the mechanistic insight that can be gained from kinetics measurements by means of a variety of examples, including unimolecular reactions, third-order reactions, enzyme reactions, chain reactions, and explosive reactions.

In closing, it is worth noting some of the aspects of reaction kinetics that we have *not* covered. Simple collision theory, covered in chapter 1, is the simplest theoretical model developed to interpret and predict chemical reaction rates. It treats the reactants as 'hard spheres', and does not include the effects of any long-range interactions—for example van der Waals or ionic interactions—or any internal degrees of freedom of the reactants. A number of much more sophisticated models are available, some of which we have alluded to at appropriate points in the text. These include transition-state theory, RRK and RRKM theory, and capture theories, all of which are discussed in more advanced texts on reaction kinetics (see section 7.6).

In focusing mainly on reactions occurring in the gas phase, molecular interactions other than those driving the reaction itself can be ignored to a very good approximation. However, for reactions in solution, particularly those involving ionic reactants, the effects of the solvent on the reaction can be significant. Reactants must diffuse together through solution before they can react, and newly-formed products find themselves confined within a 'solvent cage'. Factors such as solvent viscosity and polarity can have a dramatic effect on the reaction rate, and in some cases the solvent may play an important role in the reaction mechanism.

Reactions occurring at surfaces also deserve a mention. As well as forming the basis of many catalytic processes, this category of reactions also encompasses the field of electrochemistry. Unsurprisingly, the reaction rate can depend critically on the properties of the surface, as well as on the properties of the surrounding gas, liquid, or solution phase.

While these scenarios certainly present additional challenges, theories and models developed to describe reaction rates within these more complex environments all build on the fundamental framework outlined in the present text. Having mastered the material in this book, the reader should therefore be well equipped to explore more advanced treatments of kinetics. A few suggestions for further reading are listed in section 7.6.

7.6 Further reading

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